

Sex-specific lipidomic profiling of the marine mussel *Mytilus coruscus*: A comparative analysis of male and female lipid metabolism

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ABSTRACT

Sex-dependent lipid metabolism plays a pivotal role in physiological adaptation and reproduction in marine bivalves, yet its molecular basis remains poorly understood. Here, UPLC-HRMS-based lipidomics was applied to characterize sex-specific lipid profiles in mussels. Females showed higher levels of glycerophospholipids, particularly phosphatidylcholine and phosphatidylethanolamine enriched with polyunsaturated fatty acids, whereas males contained more glycerolipids, especially saturated fatty acid-rich triacylglycerols. Principal component analysis and orthogonal partial least squares-discriminant analysis revealed clear metabolic separation between sexes. A total of 140 differential lipids were identified, with glycerophospholipids mainly enriched in females, while glycerolipids exhibited bidirectional regulation in males. Pathway analysis indicated coordinated remodeling of glycerophospholipid metabolism, glycerolipid turnover, and polyunsaturated fatty acid-related pathways. Females also accumulated higher ceramide levels, suggesting altered sphingolipid metabolism and potential differences in lipid remodeling between sexes. These findings reveal distinct sex-dependent lipid metabolic strategies and provide insights into nutritional evaluation and precision mussel aquaculture.

1. Introduction

The marine mussel *Mytilus coruscus* (*M. coruscus*) is a commercially and ecologically important bivalve species widely distributed along the coastal regions of East Asia, particularly in the East China Sea and Yellow Sea. As a filter-feeding organism, *M. coruscus* plays a crucial role in maintaining water quality and ecosystem stability. In addition, it is considered a valuable seafood resource because of its high nutritional value, such as abundant proteins, vitamins, minerals, and bioactive lipids, especially ω -3 polyunsaturated fatty acids (PUFAs), such as eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA) (Gonçalves et al., 2021; Manz et al., 2023; Li et al., 2010). These lipid components are strongly associated with various health-promoting effects, including anti-inflammatory, cardioprotective, and neuroprotective activities (Calder, 2015). With increasing consumer awareness of health and nutrition, the demand for functional foods

containing natural bioactive compounds has grown globally. Recent studies have demonstrated that bioactive food components can exert beneficial physiological effects through the regulation of metabolic processes and disease-related pathways, highlighting the potential of natural resources for functional food development (Chen, Chen, & Fu, 2023). Marine organisms, particularly edible bivalves, represent valuable sources of functional ingredients, including polyunsaturated fatty acids (PUFAs), phospholipids, and other bioactive lipids, which contribute to their nutritional value and potential health-promoting properties.

Lipids play fundamental roles in bivalves, serving not only as energy reserves but also as essential components of cellular membranes and regulators of physiological processes, including reproduction, stress adaptation, and metabolic homeostasis (Harlina et al., 2023; Stillwell & Wassall, 2003). The mussel lipidome is highly complex and encompasses diverse classes, such as glycerolipids (GLs), glycerophospholipids (GP),

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and sphingolipids (SPs), each of which possesses distinct structural and functional properties (Cajka & Fiehn, 2014; He et al., 2020; Yang & Han, 2016). GPs are major components of cellular membranes and contribute to membrane integrity and fluidity, whereas neutral lipids, such as triacylglycerols (TGs), primarily function as energy reserves (Hu et al., 2023; Li et al., 2020). Moreover, lipid-derived metabolites, including PUFA-derived molecules and SPs, are involved in the regulation of physiological processes such as reproduction, stress responses, and cellular homeostasis (Amai et al., 2025; Han & Gross, 2003; Song et al., 2023). In marine bivalves, lipid metabolism may exhibit sex-dependent characteristics due to differences in reproductive investment and physiological demands. Females generally require substantial lipid resources, particularly phospholipids and PUFAs, to support oocyte development, embryogenesis, and early larval growth. In contrast, males allocate metabolic resources to spermatogenesis and sperm function, which may involve distinct patterns of lipid utilization and energy metabolism. Therefore, sex-specific lipid remodeling may reflect differences in reproductive demands and physiological adaptation between male and female individuals (Mittendorfer, 2005). These physiological differences suggest the sex-dependent lipid metabolism regulation may be reflected in lipid composition and metabolic pathways. Previous studies in *Mytilus* species have reported variations in lipid classes and fatty acid profiles between males and females; however, most investigations were largely limited to bulk lipid analysis and lacked molecular-level characterization (Xue et al., 2023).

Recent advances in lipidomics, particularly ultra-performance liquid chromatography coupled with high-resolution mass spectrometry (UPLC-HRMS), have enabled comprehensive profiling of lipid molecular species in complex biological systems (Han & Gross, 2003; Wang et al., 2025; Windarsih et al., 2024). This approach provides a powerful platform for investigating lipid metabolism at high resolution, revealing not only compositional differences but also metabolic reprogramming. Despite these technological advances, systematic and integrative lipidomic studies focusing on sex-dependent metabolic differences in *M. coruscus* remain limited. In particular, the coordinated remodeling of structural lipids, storage lipids, and bioactive lipid-associated metabolic processes has not yet been fully elucidated. From food and nutritional perspectives, sex-dependent lipid variation may influence the nutritional quality and targeted utilization of edible marine resources. Lipids in bivalves serve not only as energy reserves but also as important nutritional components, including phospholipids, long-chain polyunsaturated fatty acids, and other bioactive lipids that contribute to the functional value of seafood. Therefore, characterizing sex-specific lipid profiles can provide insights into nutritional differences between male and female individuals and support seafood quality evaluation and the development of high-value marine products. However, previous studies on *Mytilus* species have primarily focused on total lipid content and fatty acid composition, whereas comprehensive characterization of individual lipid species and their metabolic regulation remain limited. Thus, sex-dependent lipidomic analysis provides an effective strategy for elucidating lipid remodeling patterns and facilitating the precise utilization of edible marine resources.

Recent advances in high-resolution mass spectrometry-based omics technologies have provided powerful approaches for systematically characterizing complex metabolic alterations. Untargeted metabolomics has been successfully applied to identify global metabolic signatures and reveal potential metabolic pathways in biological systems, demonstrating its ability to uncover molecular changes beyond those detected through conventional targeted analyses (Xiang et al., 2026). Similarly, ultra-performance liquid chromatography-high-resolution mass spectrometry (UPLC-HRMS)-based untargeted lipidomics enables the comprehensive profiling of diverse lipid molecular species and provides valuable insights into lipid metabolic remodeling associated with physiological variation. The present study employed a UPLC-HRMS-based lipidomics approach to systematically characterize sex-dependent lipid metabolic reprogramming in *M. coruscus*. This study

aimed to comprehensively characterize sex-dependent lipidomic differences at the molecular level, identify key lipid signatures and potential biomarkers, and elucidate coordinated metabolic pathways associated with membrane remodeling, energy homeostasis, and lipid-mediated signaling. By integrating compositional and pathway-level analyses, this study provides new insights into the regulatory mechanisms underlying reproductive physiology and metabolic adaptation in bivalves and offers a theoretical basis for precision aquaculture and value-added utilization of marine resources.

2. Materials and methods

2.1. Reagents and chemicals

High-Performance Liquid Chromatography (HPLC)-grade methanol, acetonitrile, isopropanol, methyl tert-butyl ether (MTBE), and formic acid were purchased from Fisher Scientific (Fairfield, OH, USA). LC-MS grade ammonium acetate and deuterated lipid internal standards were obtained from Sigma-Aldrich (St. Louis, MO, USA). Ultrapure water was prepared using a Milli-Q system (Millipore, Billerica, MA, USA).

2.2. Sample preparation

M. coruscus specimens were collected from an aquaculture farm in Lianyungang, Jiangsu Province, China, on April 17, 2024. The specimens were immediately placed in an insulated foam box containing ice and transported to the laboratory within 3 h. Upon arrival, they were classified as male or female based on the color and morphological characteristics of the gonads. Gonadal tissues were carefully dissected, immediately frozen in liquid nitrogen, and stored at -80°C until lipidomic analysis. Four biological replicates were included for each sex, with each replicate prepared and analyzed independently. All specimens were collected from the same aquaculture area at the same time to minimize potential seasonal and environmental variations.

2.3. Lipid extraction

Approximately 50 mg (50 ± 5 mg) of mussel gonadal tissue was weighed into a 2 mL centrifuge tube containing a 6 mm grinding bead. Subsequently, 280 μL of methanol-water (2:5, v/v) and 400 μL of methyl tert-butyl ether (MTBE) were added. The samples were homogenized at low temperature for 6 min (-10°C , 50 Hz), followed by ultrasonic-assisted extraction for 30 min (5°C , 40 kHz). After incubation at -20°C for 30 min, the samples were centrifuged at $13,000 \times g$ for 15 min at 4°C . Approximately 350 μL of the supernatant was collected and evaporated to dryness under a gentle stream of nitrogen. The dried extracts were reconstituted in 100 μL of isopropanol-acetonitrile (1:1, v/v) and subjected to LC-MS/MS analysis. Quality control (QC) samples were prepared by pooling equal volumes of all the extracts and analyzing them every 5 samples to monitor system stability.

2.4. LC-MS analysis

Liquid Chromatograph Mass Spectrometry (LC-MS) analysis was performed according to previously reported methods (Hu et al., 2023; Liebisch et al., 2013; Peng et al., 2024). Chromatographic separation was performed using a Thermo Scientific Vanquish Ultra High-Performance Liquid Chromatography (UHPLC) system equipped with an Accucore C30 column (100×2.1 mm, $2.6 \mu\text{m}$; Thermo Fisher Scientific, USA) maintained at 40°C . The mobile phase consisted of (A) 50% acetonitrile/water with 0.1% formic acid and 10 mmol/L ammonium acetate and (B) acetonitrile/isopropanol/water (10/88/2, v/v/v) containing 0.02% formic acid and 2 mmol/L ammonium acetate. A gradient elution was applied at a flow rate of 0.3 mL/min: 0–2 min, 30% B; 2–15 min, 30–50% B; 15–25 min, 50–99% B; 25–30 min, 99% B; 30–32 min, 99–30% B; and 32–35 min, 30% B. The injection volume was

3 μ L.

Mass spectrometry was conducted using a Q Exactive HF-X hybrid quadrupole-Orbitrap mass spectrometer (Thermo Fisher Scientific) in both positive (ESI+) and negative (ESI-) ionization modes. The key parameters included the spray voltage (± 3000 V), sheath gas flow (60 psi), auxiliary gas flow (20 psi), ion transfer tube temperature (370 °C), and full-scan range (m/z 200–2000). Data-dependent tandem mass spectrometry (MS/MS) scans were acquired with normalized collision energies (20–60%) and a resolution of 15,000.

2.5. Lipid identification and quantification

Raw LC-MS/MS data were imported into LipidSearch software (Thermo Fisher Scientific, USA) for lipid identification and relative quantification. A signal-to-noise ratio (S/N) threshold of ≥ 3 was applied, and only lipid features meeting this criterion were retained. Retention times were aligned using QC samples to correct for deviations among analytical runs. Preliminary lipid annotation was performed using MS1 precursor-ion information, including m/z values, adduct types, and isotopic patterns, with a mass tolerance of 10 ppm. Lipid identities were further confirmed using MS/MS fragmentation information, including fragment ions, collision energies, and fragment-

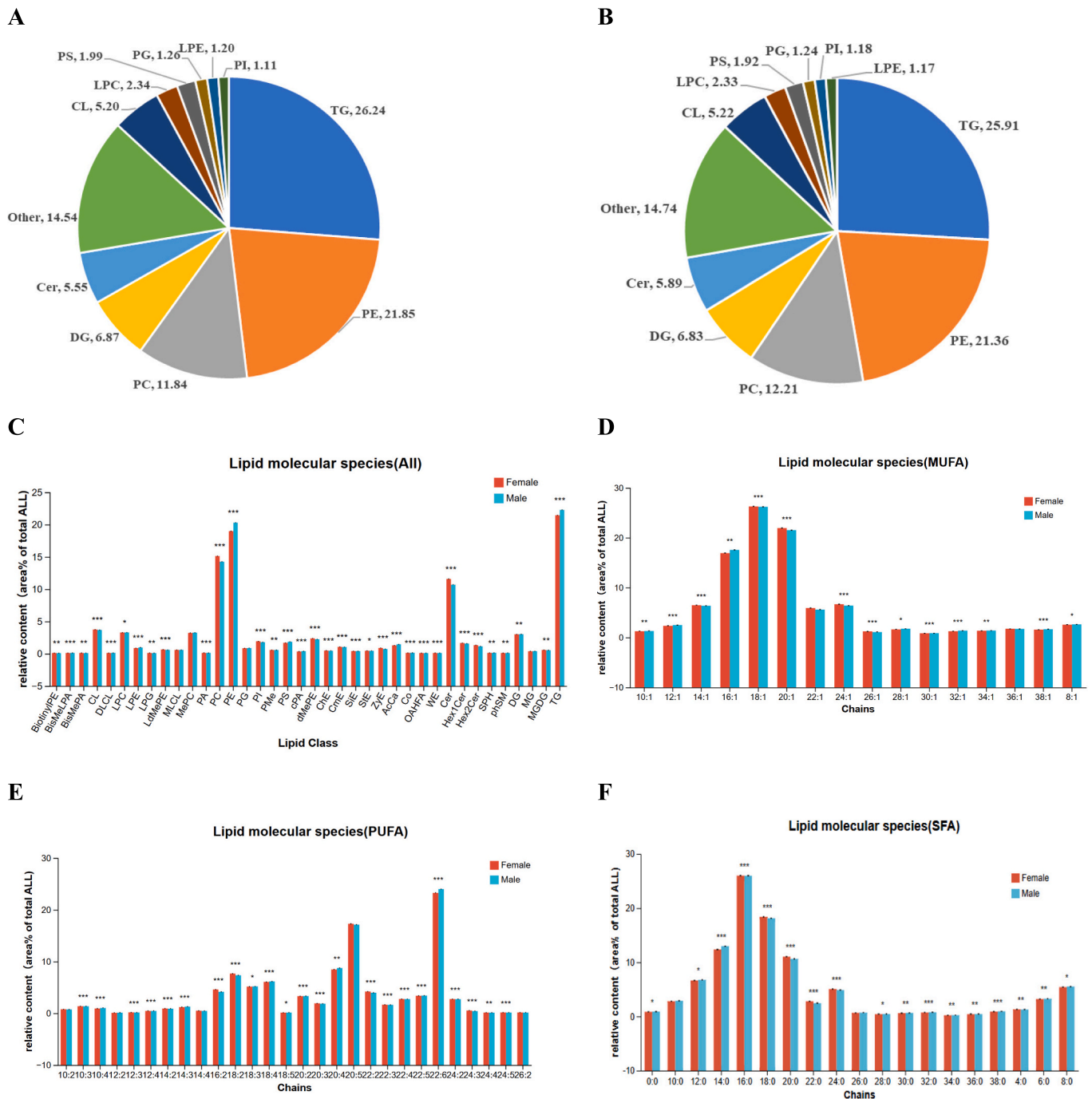


Fig. 1. Lipid class composition of male (A) and female (B) *M. coruscus*. Comparative analysis of lipid composition and fatty acid profiles between male and female *M. coruscus*. Distribution of major lipid subclasses (C). Relative composition of monounsaturated fatty acids (MUFAs) (D). Relative composition of polyunsaturated fatty acids (PUFAs) (E). Relative composition of saturated fatty acids (SFAs) (F). The data are expressed as relative percentages.

intensity patterns. When multiple ion features were assigned to the same lipid species, the ion exhibiting the highest identification confidence and suitable quantitative performance was selected. Relative lipid abundance was determined by chromatographic peak-area integration. Lipid features with missing values in >20% of the samples were removed, and the remaining missing values were imputed using the minimum-value method. The resulting data were normalized to the total peak intensity of each sample. Features with a relative standard deviation (RSD) > 30% of the QC samples were excluded before subsequent statistical analysis.

2.6. Statistical analysis

Multivariate statistical analyses, including principal component analysis (PCA) and orthogonal partial least-squares discriminant analysis (OPLS-DA), were performed using R software. Differentially abundant lipids between male and female mussels were identified based on the following criteria: variable importance in projection (VIP) > 1.0, fold change (FC) > 1.5, and Student's *t*-test ($p < 0.05$). Four independent biological replicates per sex were included in the statistical analyses. Figures were generated using GraphPad Prism.

3. Results and discussion

3.1. Lipid class composition differs between sexes

A comparative analysis of lipid composition revealed pronounced sex-dependent differences in *M. coruscus*. As shown in Fig. 1, GLs and GPs were the dominant lipids in both groups, collectively contributing to more than 80% of the total lipid content. Females exhibited a higher proportion of GPs, whereas males showed relatively greater abundance of GLs. In contrast, SPs and other minor lipid classes accounted for less than 10% of the total lipids and showed relatively limited variation between the sexes. The lipid-subclass composition differed between male and female *M. coruscus*. GPs were the predominant lipid subclass in both sexes, accounting for approximately 40% and 37% of the total identified lipids in males and females, respectively. Differences were also observed in the relative proportions of GLs and other lipid subclasses, indicating sex-specific variation in gonadal lipid composition. This lipid redistribution suggests that sex-dependent lipid metabolism reflects not only quantitative differences but also functional specialization between structural lipids and storage lipids (Li et al., 2020). Similar lipidomic patterns have been reported in shellfish and fish, where GLs and TGs were typically the dominant lipid subclasses (Cao et al., 2024; Gaspar et al., 2023; Liu et al., 2026). In a lipidomic analysis of four edible shellfish species, the proportion of PLs varied markedly across species, ranging from 29.6% to 54.9% (Li et al., 2020; Nava et al., 2023; Sun et al., 2021). PE consistently emerged as the dominant PL subclass across all four species, while TGs represented the most abundant lipid class in *Ostrea gigas* (39.1%) and *M. edulis* (27.5%). Collectively, these findings indicate that GPs and GLs together constitute the principal lipid framework in aquatic organisms (Li et al., 2020). In the present study, the enrichment of GPs in females may reflect increased demand for membrane construction, and oocyte development, whereas the higher GL proportion in males suggests greater reliance on neutral lipid reserves for energy mobilization.

As shown in Fig. 1C, Phosphatidylcholine (PC), phosphatidylethanolamine (PE), and TG were the predominant subclasses in both groups. Females exhibited higher proportions of PC and PE, whereas TGs were more enriched in males. Fatty acid profiling further revealed distinct compositional patterns between sexes. PUFAs were significantly elevated in females, whereas SFAs were more abundant in males (Fig. 1E–F). In contrast, MUFAs showed relatively minor differences between groups (Fig. 1D). PCs are predominant structural phospholipids and play an essential role in maintaining permeability and structural integrity, whereas TGs act as major storage lipids, energy source, and fatty acid carriers. Therefore, the higher abundance of PCs and PEs in

females may indicate enhanced membrane phospholipid biosynthesis and remodeling (Jeon et al., 2025; Minami et al., 2026). Fatty acid profiling further revealed distinct patterns. In addition, the higher PUFA levels in females may contribute to membrane fluidity and lipid compositional adaptation, whereas the increased SFA content in males may favor lipid stability and long-term energy retention (Calder, 2015). Similar fatty acid distribution patterns have been reported in aquatic products, where DHA (C22:6 n-3), EPA (C20:5 n-3), palmitic acid (C16:0), stearic acid (C18:0), oleic acid (C18:1 n-9), and linoleic acid (C18:2 n-6) are among the predominant fatty acids detected in seafood species (Nava et al., 2023).

3.2. Multivariate analysis reveals clear separation of lipidomic profiles

To comprehensively assess the global lipidomic variation, PCA and OPLS-DA analyses were performed. As shown in Fig. 2A–B, male and female individuals were clearly separated along the principal components, with PC1 explaining approximately 60–70% of the total variance, indicating pronounced intrinsic metabolic differences between the sexes. The tight clustering of samples within each group further suggests good reproducibility and strong intragroup consistency. OPLS-DA further improved group discrimination, with complete separation between the male and female groups and no obvious overlap observed (Fig. 2C–D). The predictive component accounted for most of the variation, whereas the orthogonal component contributed limited, indicating that the variation was predominantly driven by sex-related differences rather than random noise.

3.3. Differential lipid species highlight distinct metabolic characteristics

Volcano plot analysis was performed to identify significantly altered lipid species between sexes (Fig. 2E). The distribution of lipids differed markedly between upregulated and downregulated molecules, indicating extensive metabolic differences. Many lipid species met the screening criteria (VIP > 1, $p < 0.05$, FC > 1.5). Most upregulated lipids in females belonged to glycerophospholipid subclasses, particularly PCs and PEs, whereas many downregulated lipids were associated with GLs, especially TG species. The magnitude of change varied considerably across lipid species, with several exhibiting FC values greater than 1, indicating more than twofold differences between groups. In addition, larger point sizes were predominantly observed among significantly altered lipids, suggesting that these molecules contributed substantially to group discrimination in the OPLS-DA model. Overall, the volcano plot highlights a broad and coordinated shift in lipid metabolism, with females showing enrichment of structural lipids and males exhibiting a relative depletion of these components.

To further characterize the distribution patterns and identities of differential lipids, hierarchical clustering and lipid annotation analyses were performed. As shown in the heatmap (Fig. 3A), the male and female samples formed two clearly distinct clusters, indicating consistent intragroup similarity and pronounced intergroup variation. Lipids within the same class tended to cluster together, particularly GPs, which were predominantly upregulated in females, whereas GLs displayed more heterogeneous patterns. The accompanying scatter plot (Fig. 3B) further confirmed a wide dynamic range of lipid variation, with numerous species exhibiting distinct abundance shifts between male and female groups.

As summarized in Table 1, a total of 140 differentially lipids were identified between male and female *M. coruscus*, including 80 triacylglycerols, 26 phosphatidylcholines, 13 diacylglycerols (DGs), 10 phosphatidylethanolamines, 5 sphingomyelins (SMs), 3 lysophosphatidylcholines (LPCs), 2 phosphatidylinositols (PIs), and 1 ceramide (Cer). The predominance of TG species among the differential lipids indicates that glycerolipid metabolism is highly responsive to physiological regulation and may play a central role in sex-dependent lipid remodeling (Cui et al., 2024; Ying et al., 2024; Zhang et al., 2025). Several TG

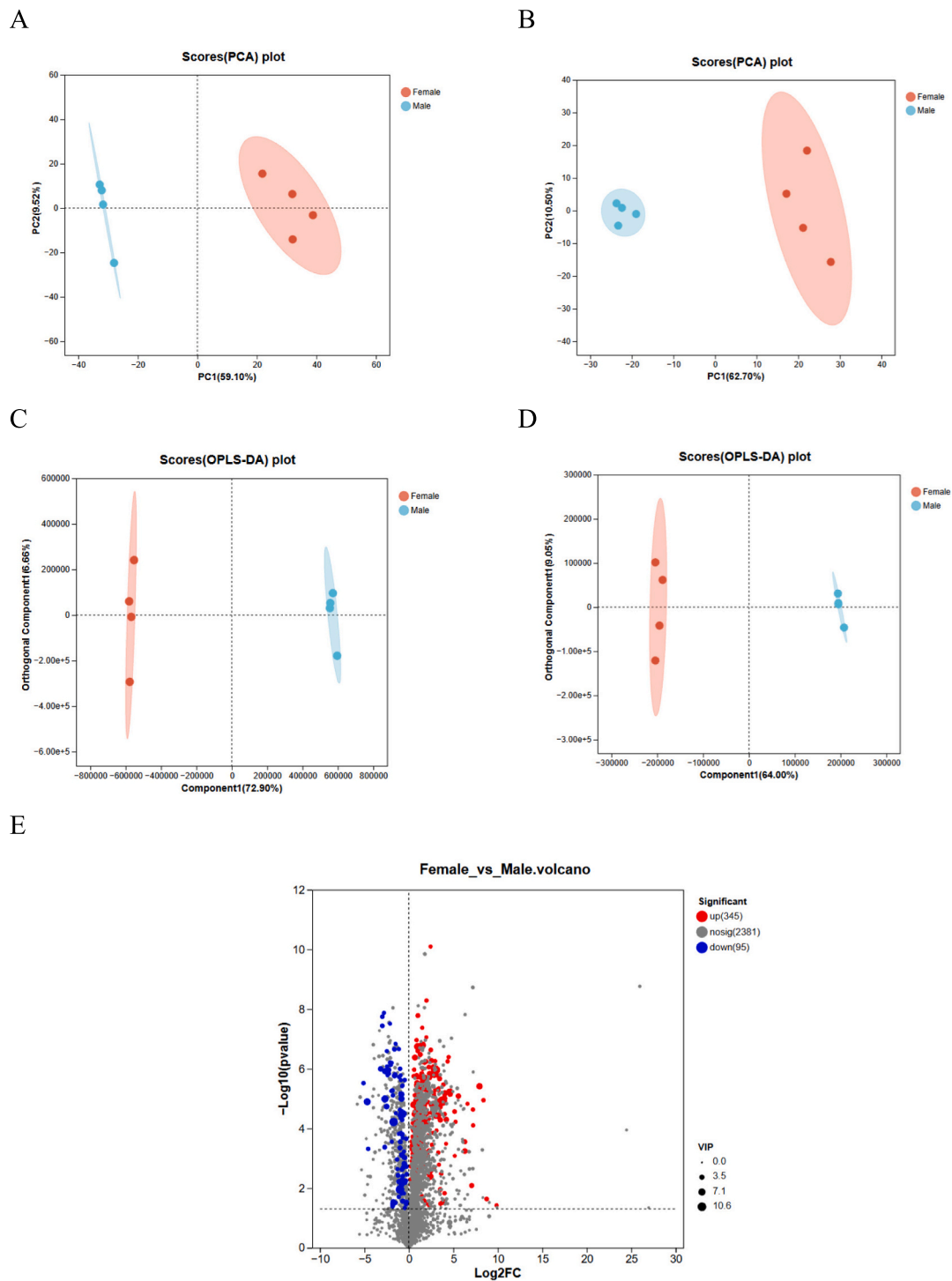


Fig. 2. Multivariate statistical analysis of lipidomic profiles in male and female *M. coruscus*. Principal component analysis (PCA) score plot in negative ion mode (A). PCA score plot in positive ion mode (B). Orthogonal partial least squares-discriminant analysis (OPLS-DA) score plot in negative ion mode (C). OPLS-DA score plot in positive ion mode (D). Volcano plot of differential lipids between male and female *M. coruscus* (E).

species, such as TG (15:0/16:0/18:3) and TG (20:5/20:5/20:5), were significantly upregulated, suggesting enhanced lipid storage or mobilization. In contrast, other species, such as TG (16:1/22:6/22:6), were markedly downregulated, indicating the selective utilization of fatty acid chains. This bidirectional variation observed in TG species further suggests dynamic lipid turnover and selective fatty acid partitioning. TGs can be hydrolyzed into DGs and free fatty acids, which may subsequently enter into β -oxidation or phospholipid biosynthesis (Farese & Walther, 2009).

In contrast to TG, glycerophospholipid, particularly PC species, were more consistently upregulated in females. Representative molecules such as PC (20:0e/18:1) and PC (18:0e/22:2) were significantly increased. PCs are essential components of cellular membranes and serve as reservoirs for bioactive PUFAs, and their enrichment suggests enhanced membrane remodeling, lipid trafficking, and metabolic processes (Calder, 2015; Cao et al., 2025; Song, et al., 2024; Van Meer et al., 2008). Similarly, the PE species exhibited moderate but coordinated changes. Notably, sphingolipid metabolism clearly exhibited sex-

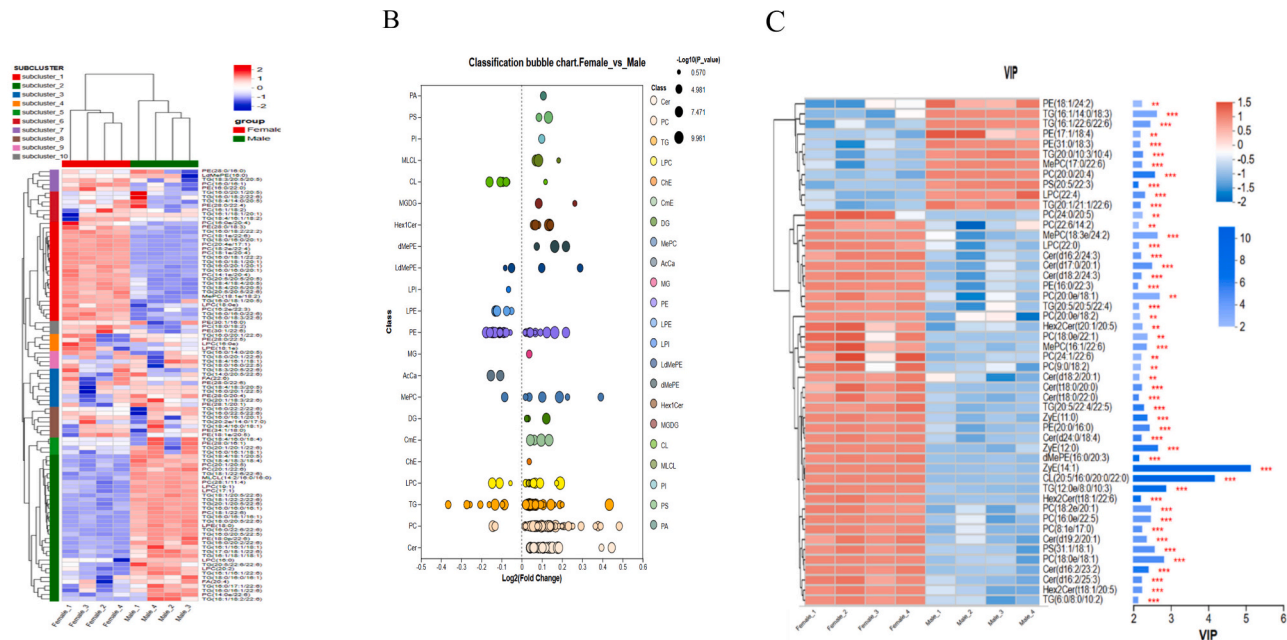


Fig. 3. Clustering analysis (A) and scatter distribution (B) of differential lipids for *M. coruscus*. Heatmap of the top 50 differential lipids between male and female *M. coruscus* (C).

Table 1
Differential lipid species between male and female *M. coruscus*.

Lipid	Category	Class	FC	P_value	Lipid	Category	Class	FC	P_value
TG(15:0/16:0/18:3)	GL	TG	1.14	0.0002217	PE(18:0e/20:5)	GP	PE	1.07	0.001658
TG(6:0/11:2/14:3)	GL	TG	1.07	0.0001257	PE(29:1/18:2)	GP	PE	1.03	0.00652
TG(20:5/14:1/20:5)	GL	TG	1.06	0.00839	PE(28:1/20:5)	GP	PE	1.02	0.0003046
TG(20:5/20:5/20:5)	GL	TG	1.03	0.0001666	PE(14:1e/11:1)	GP	PE	1.01	0.000628
TG(18:0/20:1/22:2)	GL	TG	1.02	0.0001386	PE(9:0/22:0)	GP	PE	1.01	0.00685
TG(18:4/18:4/20:5)	GL	TG	1.02	0.0004875	PE(17:0/22:6)	GP	PE	0.93	0.0001141
TG(18:0e/16:0/22:2)	GL	TG	1.02	0.003241	PC(20:0e/18:1)	GP	PC	1.31	0.001142
TG(18:3e/10:3/10:3)	GL	TG	1.02	0.00581	PC(18:0e/22:2)	GP	PC	1.29	0.0057
TG(20:1/18:1/22:2)	GL	TG	1.02	0.0002958	PC(16:2e/21:0)	GP	PC	1.19	0.0096
TG(20:1/18:1/20:2)	GL	TG	1.01	0.0004192	PC(20:0e/18:2)	GP	PC	1.18	0.0074
TG(20:5/20:5/22:6)	GL	TG	1.01	0.001038	PC(18:1e/18:2)	GP	PC	1.16	0.002076
TG(16:1/16:1/20:3)	GL	TG	1.01	0.001908	PC(16:2e/23:0)	GP	PC	1.15	0.0057
TG(18:4/16:1/17:1)	GL	TG	0.94	0.004075	PC(20:0e/20:4)	GP	PC	1.12	0.00407
TG(17:0/20:5/22:6)	GL	TG	0.87	0.000886	PC(20:1e/22:6)	GP	PC	1.11	0.000588
TG(16:1/22:6/22:6)	GL	TG	0.83	0.0001215	PC(16:2e/22:3)	GP	PC	1.11	0.000957
TG(16:1/14:0/18:3)	GL	TG	0.78	0.0003577	PC(20:4e/19:1)	GP	PC	1.11	0.000868
PS(18:4/19:0)	GP	PS	1.06	0.0002353	PC(20:1/20:4)	GP	PC	1.10	0.001608
MG(22:6)	GL	MG	1.03	0.0003522	PC(20:0/22:6)	GP	PC	1.10	0.00617
MePC(18:3e/24:2)	GP	MePC	1.31	0.0003671	PC(18:4/16:1)	GP	PC	1.06	0.0002353
MePC(20:1e/18:2)	GP	MePC	1.17	0.00869	PC(20:5/22:6)	GP	PC	1.06	0.0003389
MePC(18:1e/18:2)	GP	MePC	1.03	0.0001224	PC(16:1/20:5)	GP	PC	1.06	0.0004875
MePC(18:0e/22:6)	GP	MePC	1.01	0.004458	PC(18:0/18:3)	GP	PC	1.06	0.001704
LPE(20:5)	GP	LPE	0.97	0.00184	PC(22:6/22:6)	GP	PC	1.05	0.0001553
LPC(18:1e)	GP	LPC	1.03	0.001418	PC(16:2e/19:0)	GP	PC	1.04	0.0004367
LPC(20:0)	GP	LPC	1.02	0.001137	PC(18:0/20:5)	GP	PC	1.04	0.0002635
LPC(22:2)	GP	LPC	1.01	0.000628	PC(16:1/14:0)	GP	PC	1.03	0.0002161
LdMePE(14:0)	GP	LdMePE	1.22	0.0002451	PC(16:1/23:0)	GP	PC	1.02	0.0001946
Hex1Cer(t20:0/22:6)	SP	Hex1Cer	1.06	0.000892	PC(10:1e/11:4)	GP	PC	1.02	0.001042
dMePE(16:0e/20:5)	GP	dMePE	1.05	0.000755	PC(16:0/21:0)	GP	PC	1.02	0.000436
DG(18:4/20:5)	GL	DG	1.02	0.00581	PC(16:1e/18:2)	GP	PC	1.02	0.0004214
DG(18:4/22:6)	GL	DG	1.02	0.00501	PC(16:0/16:1)	GP	PC	1.01	0.004995
CL(17:0/20:1/20:1/22:6)	GP	CL	0.95	0.00747	PC(18:2/22:6)	GP	PC	1.01	0.004089
Cer(d18:2/20:1)	SP	Cer	1.32	0.001645	PC(16:0/12:0)	GP	PC	1.01	0.00789
Cer(d16:1/22:0)	SP	Cer	1.14	0.001697	PA(26:0/22:6)	GP	PA	1.08	0.0001057

Notes: FC (fold change) represents the relative abundance ratio between the female and male groups (female/male). Lipids with a FC > 1 are upregulated, whereas those with a FC < 1 are downregulated. Statistical significance was determined using Student's *t*-test (*p* < 0.05). Only representative differential lipids are shown here; the full dataset is provided in the Supporting Information. Lipid categories: GL, GLs; GP, GPs; SP, SPs.

dependent regulation, as indicated by the significant accumulation of ceramide species, particularly Cer (d18:2/20:1), in females. Ceramides are bioactive lipids involved in apoptosis, differentiation, and stress

signaling (Han & Gross, 2003; Maceyka & Spiegel, 2014). In addition, minor lipid subclasses, such as LPC, LPE, and DG, also exhibited significant abundance changes, reflecting altered phospholipid turnover,

membrane remodeling, and intermediate lipid metabolism.

3.4. Heatmap analysis of key differential lipids

The heatmap of the top 50 differential lipids provided an overall view of sex-dependent lipid variation patterns. As shown in Fig. 3C, the male and female samples formed two distinct clusters. GPs, especially PCs and PEs, were predominantly upregulated in females, whereas several TG species were more abundant in males. This distribution pattern was consistent with the lipid class analysis, fatty acid composition, and differential lipid screening, indicating strong agreement across multiple analytical levels. In addition, lipids belonging to the same subclass tended to cluster together, indicating coordinated regulation within lipid classes. In contrast, GLs exhibited more dispersed patterns, reflecting greater metabolic flexibility and turnover.

Similar heatmap-based clustering approaches have been widely used in aquatic lipidomic studies to visualize lipid remodeling under different physiological or processing conditions (Cajka et al., 2017; Duan et al., 2022). For example, Zhong et al. reported distinct clustering of lipid profiles during the tilapia muscle steaming process, in which lipid distribution patterns varied substantially over processing time (Sun et al., 2021). In the present study, the heatmap analysis suggests that sex-dependent lipid remodeling lipid variation involves coordinated metabolic remodeling rather than isolated lipid changes.

3.5. Comparison of representative lipid subclasses

Lipid remodeling is associated not only with energy storage but also with the maintenance of membrane integrity and the regulation of cellular responses to environmental and physiological challenges. Recent studies have demonstrated that specific lipid compositions can serve as protective barriers against oxidative stress by modulating membrane properties and redox sensitivity (Ma et al., 2026). The comparison of representative differential lipid species between male and female *M. coruscus* is shown in Fig. 4. Significant sex-dependent differences were observed across multiple lipid classes. For TGs, a clear bidirectional pattern was observed. Several PUFA-rich TG species, such as TG (16:0/18:1/20:5) and TG (20:5/20:5/20:5), were significantly enriched in females, whereas TG (16:1/22:6/22:6) and TG (17:0/20:5/22:6) showed markedly higher abundance. These results indicate the selective accumulation and utilization of specific fatty acid combinations rather than uniform TG regulation (Chen, et al., 2023; Song, et al., 2024). DG species also exhibited distinct sex-dependent variation. DG (21:1/11:3) was significantly increased in females, whereas DG (16:0/18:3) was more abundant in males. Because DGs serve as key intermediates linking TG metabolism and phospholipid biosynthesis, their divergent regulation suggests sex-specific differences in lipid remodeling pathways. Ceramide species were consistently upregulated in females. In particular, Cer (d16:0/22:0) and Cer (d16:1/19:0) displayed pronounced female-biased enrichment, suggesting that sphingolipid metabolism may be more active in females. Although SPs are relatively underexplored in bivalve lipidomics, they are recognized as important roles in cell signaling and membrane organization. Therefore, the elevated ceramide enrichment observed in females may be associated with oocyte maturation, cellular differentiation, or membrane remodeling.

PC species exhibited the most consistent female-biased enrichment. Representative PUFA-containing PCs, such as PC (16:0/22:5) and PC (19:1/20:5) were significantly elevated in females. PCs are major structural phospholipids that contribute to membrane integrity and serve as reservoirs for long-chain PUFAs, such as EPA and DHA. Previous lipidomics studies have reported that PCs are major phospholipid in different aquatic products. In the hepatopancreas of *Penaeus vannamei*, 28.21% PCs and 11.60% PEs were detected among 1699 lipids of 32 classes by LC-MS/MS (Duan et al., 2022). Additionally, the relative contents of PCs in the head, meat, and viscera of Pacific saury were

shown to be 10.90%, 7.38%, and 6.00%, respectively (Tao et al., 2024). Therefore, the enrichment of PC species in females may indicate enhanced membrane remodeling and altered phospholipid homeostasis.

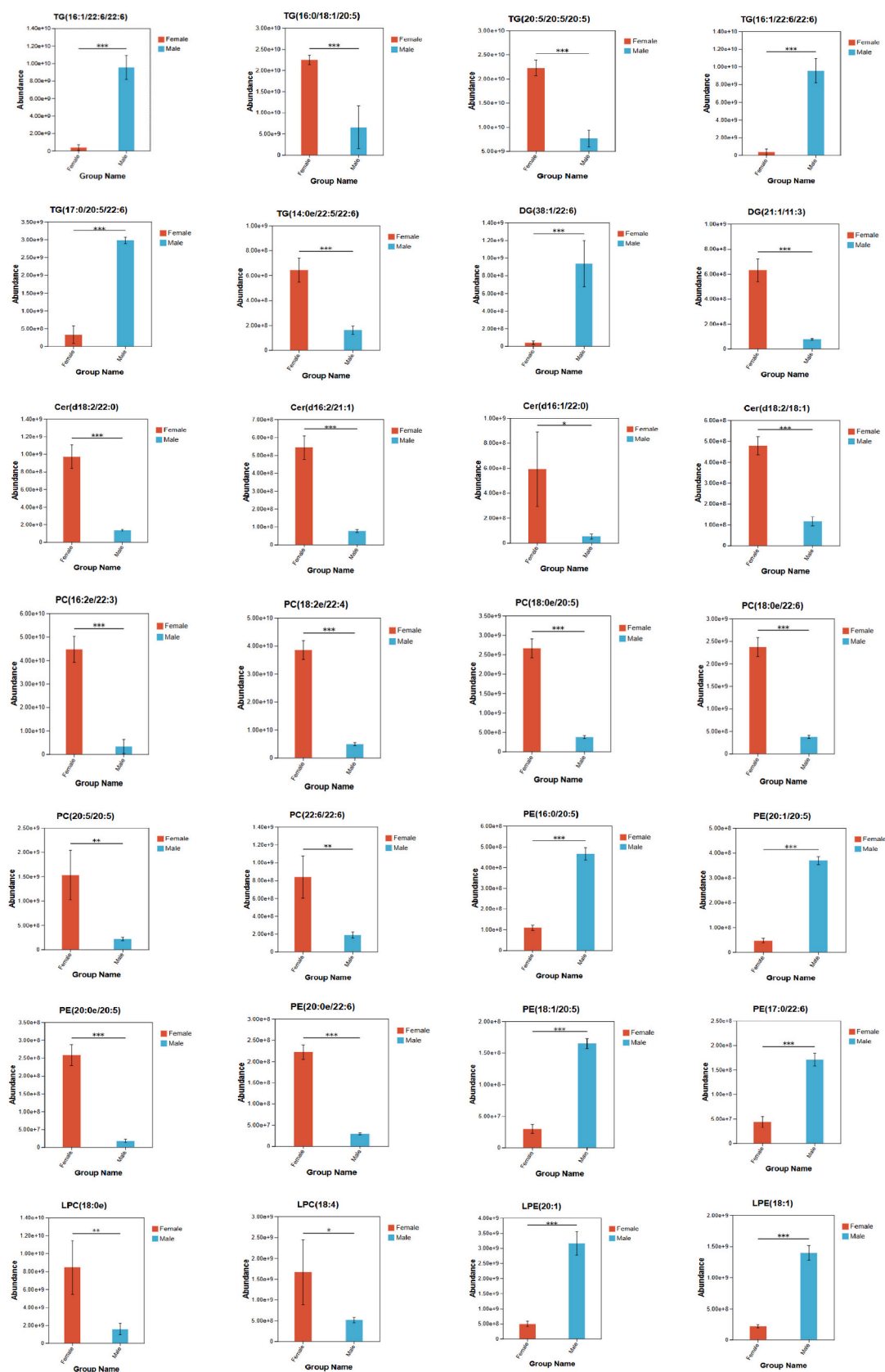
PEs exhibited heterogeneous distribution patterns. Some PEs, such as PE (18:1/22:6), were elevated in females, whereas others showed higher abundance in males. This heterogeneity suggests that PE subclass may possess specific functions depending on fatty acyl composition. Because PEs are closely involved in membrane curvature, organelle dynamics, and autophagy-related membrane processes. Therefore, the selective remodeling of PEs may reflect differences in membrane dynamics and intracellular turnover between sexes.

Lysophospholipid levels, particularly those of LPCs, were also elevated in females, suggesting enhanced phospholipid turnover. LPCs are generated through phospholipid hydrolysis and reacylation processes and are commonly considered indicators of membrane remodeling (Liu et al., 2023; Zhou et al., 2023). Previous lipidomics studies have reported increased LPC accumulation during phospholipid remodeling in aquatic products (Shui et al., 2025). In the present study, however, the simultaneous enrichment of LPC, PC, and PE species suggests regulated phospholipid turnover rather than passive lipid degradation. In the present study, sex-dependent differences in structural phospholipids and ceramide-related lipids suggest that membrane lipid remodeling may represent an adaptive strategy for maintaining cellular homeostasis and meeting the distinct physiological demands of male and female *M. coruscus*.

3.6. KEGG pathway enrichment analysis

To further elucidate the metabolic networks associated with sex-dependent lipid variation, KEGG pathway enrichment analysis was performed at both level 2 and level 3 (Fig. 5). Differential lipids were extensively distributed across multiple interconnected pathways. Among these, glycerophospholipid metabolism was the most significantly enriched pathway and involved a large number of lipid species, including PCs, PEs, LPCs, and PSs. Representative molecules such as PC (22:6/22:6), PC (20:1/20:4), LPC (20:4), and PS (18:0/22:6) were widely distributed within this pathway, indicating extensive membrane lipid remodeling (Amai et al., 2025; Cajka & Fiehn, 2014; Hu et al., 2009; Xue et al., 2023). In particular, the enrichment of PUFA-rich PC species in females suggests enhanced membrane fluidity and adaptive structural regulation, suggesting the pivotal role of PUFA-associated phosphatidylcholine in sex-dependent lipid metabolism. In addition, arachidonic acid metabolism and other PUFA-related pathways, including α -linolenic acid and linoleic acid metabolism, were mainly associated with PC molecules, such as PC (18:3/20:5), PC (22:5/22:6), and PC (16:0/20:3). These results indicated that these pathways are functionally interconnected and collectively contribute to PUFA metabolism and lipid metabolic adaptation. Given that phospholipid-bound PUFAs can serve as precursors for bioactive lipid mediators, the coordinated enrichment of these lipids in females may suggest altered PUFA-associated lipid metabolism, which is consistent with the elevated PUFA levels and phospholipid abundance observed in female mussels.

Autophagy-related pathways and GPI-anchor biosynthesis were primarily associated with PE species, particularly PE (14:0/22:6) and PE (20:0/16:0), suggesting a potential role for specific phospholipids in membrane dynamics, vesicle formation, and protein anchoring processes. Moreover, glycine, serine, and threonine metabolism were linked to PS (18:0/22:6), highlighting a connection between phospholipid metabolism and amino acid-related pathways. This linkage suggests that lipid metabolism is closely integrated with broader cellular metabolic networks, including nitrogen metabolism and biosynthetic pathways, thereby contributing to cellular homeostasis and physiological regulation. In terms of energy metabolism, glycerolipid metabolism is mainly represented by TG species, such as TG (18:0/18:0/18:0), reflecting the regulation of neutral lipid storage and mobilization. These findings are consistent with the extensive variation and bidirectional regulation of

Fig. 4. Relative abundance of major lipid subclasses in male and female *M. coruscus*.

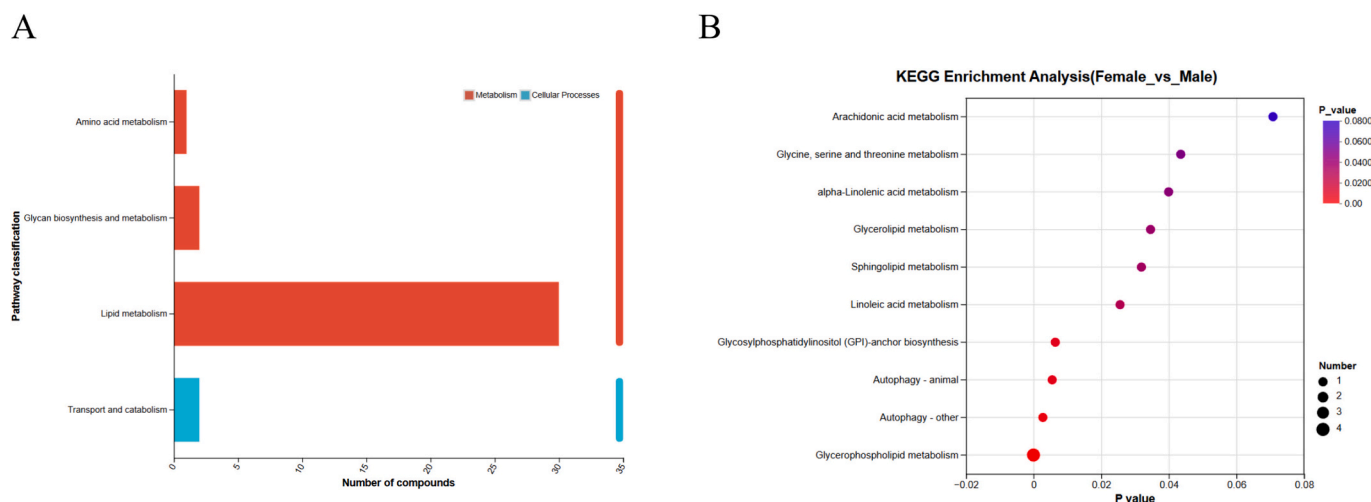


Fig. 5. KEGG pathway enrichment analysis of differential lipids at levels 2 (A) and 3 (B) for *M. coruscus*.

TG species, supporting the role of TG as dynamic lipid reservoirs. Rather than serving solely as passive energy storage molecules, TG species may participate in selective fatty acid redistribution between structural and stored lipid pools in response to sex-specific metabolic demands (Martakos et al., 2026; Ying et al., 2024; Zhang et al., 2025).

Furthermore, sphingolipid metabolism is characterized by the enrichment of multiple ceramide species, including Cer (d18:1/16:0), Cer (d18:1/18:0), and Cer (d18:1/24:0), the variation in ceramide-related lipids suggests potential alterations in sphingolipid metabolism and membrane-associated lipid regulation, which may contribute to physiological adaptation. Ceramides act as key bioactive lipids that regulate apoptosis, stress responses, and cellular differentiation. Although ceramide accumulation has previously been associated with oxidative stress in processed aquatic products, the consistent enrichment observed in females in this study is more likely related to regulated physiological processes, such as reproductive-associated membrane remodeling and cellular signaling activity. Lipid metabolic pathways, particularly glycerolipid metabolism, are essential for regulating lipid storage, turnover, and energy homeostasis. In the present study, glycerolipid metabolism was significantly enriched among pathways associated with differential lipids, suggesting that sex-dependent lipid remodeling in *M. coruscus* may involve differences in lipid accumulation, mobilization, and energy allocation between males and females. Previous studies have shown that natural functional components can regulate glucose and lipid metabolism through structure-dependent metabolic modulation, highlighting the complex interactions between bioactive molecules and metabolic networks (Jia et al., 2026; Dou et al., 2024). Therefore, the enrichment of glycerolipid-related pathways observed in this study may reflect the coordinated regulation of lipid metabolism associated with physiological differences between the sexes.

Sex-dependent lipidomic differences in *M. coruscus* likely reflect distinct physiological demands associated with reproductive investment, energy allocation, and membrane homeostasis. Variations in GPs, GLs, and SPs may indicate differential regulation of membrane structure, lipid storage, and metabolic adaptation between males and females. These findings provide insights into the biological basis of lipid remodeling during bivalve reproduction. As an edible marine resource, the lipid composition of *M. coruscus* also contributes to its nutritional characteristics and potential value as a functional food. Therefore, understanding sex-dependent lipid variation may provide useful information for seafood quality evaluation and the targeted utilization of marine resources.

Environmental and reproductive factors are important drivers of lipid metabolism in filter-feeding bivalves. Variations in food availability, feeding activity, and water temperature may influence lipid

accumulation, energy storage, and metabolic adaptation, whereas changes in gonadal maturation and spawning cycles can alter lipid allocation between reproductive investment and somatic maintenance. In this study, male and female *M. coruscus* individuals were collected from the same aquaculture site during the same period and classified according to gonadal characteristics, thereby minimizing potential environmental and reproductive variation between the groups. Nevertheless, detailed assessments of gonadal maturation stages and continuous environmental monitoring were not performed, and the potential interactions between ecological factors and sex-dependent lipid remodeling require further investigation. Future studies integrating multiseason sampling, environmental measurements, reproductive histology, and multi-omics approaches will provide deeper insights into the ecological and physiological regulation of lipid metabolism in bivalves.

4. Conclusion

In summary, this study provides a comprehensive lipidomic characterization of sex-dependent metabolic differences in *M. coruscus*, revealing coordinated lipid metabolic patterns between males and females. Females were characterized by elevated levels of GPs and polyunsaturated fatty acids, reflecting differences in membrane remodeling and lipid metabolic adaptation, whereas males exhibited a lipid profile enriched in neutral lipids, indicating a metabolism oriented toward energy storage. Integrated pathway analysis further demonstrated that these differences are driven by interconnected metabolic networks involving glycerophospholipid remodeling, triglyceride turnover, and sphingolipid metabolism. Overall, these findings provide new insights into sex-dependent lipid remodeling and nutritional characteristics in bivalves, supporting improved nutritional evaluation, precision aquaculture, and the targeted development of high-value marine food products.

CRediT authorship contribution statement

Lipin Chen: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Yu Liu:** Writing – review & editing, Methodology. **Gang Yu:** Writing – review & editing, Resources, Investigation. **Mingfei Pan:** Software, Resources, Methodology, Investigation. **Guanghua Xia:** Supervision, Resources. **Zhaojie Li:** Software, Resources. **Haohao Shi:** Writing – review & editing, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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